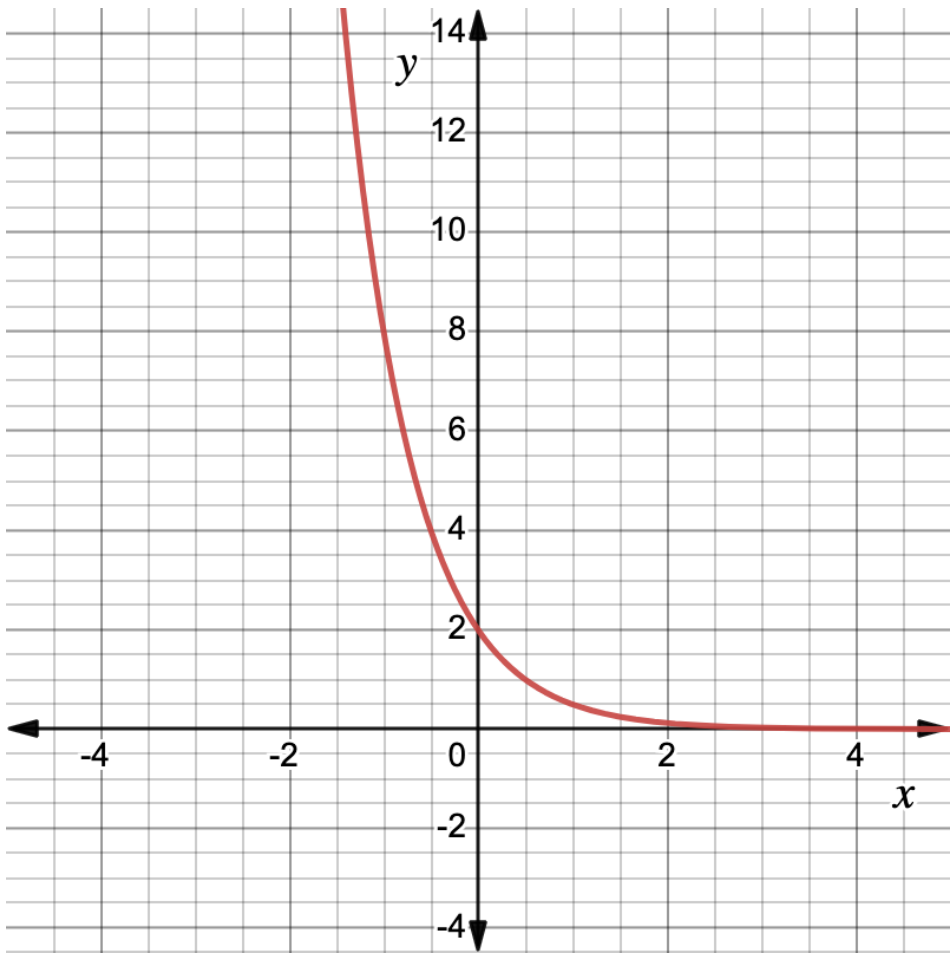


Forms of Exponential Functions	$y = ab^x$	$y = a(1 \pm r)^x$
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1. An equation is shown.

$$y = 2\left(1 - \frac{3}{4}\right)^x$$

a. Use the coordinate grid to graph the equation.



b. Complete the table.

Name	Key Feature	Name	Key Feature
x -intercept(s)	none	increasing	none
y -intercept	(0, 2)	decreasing	$\mathbb{R}$

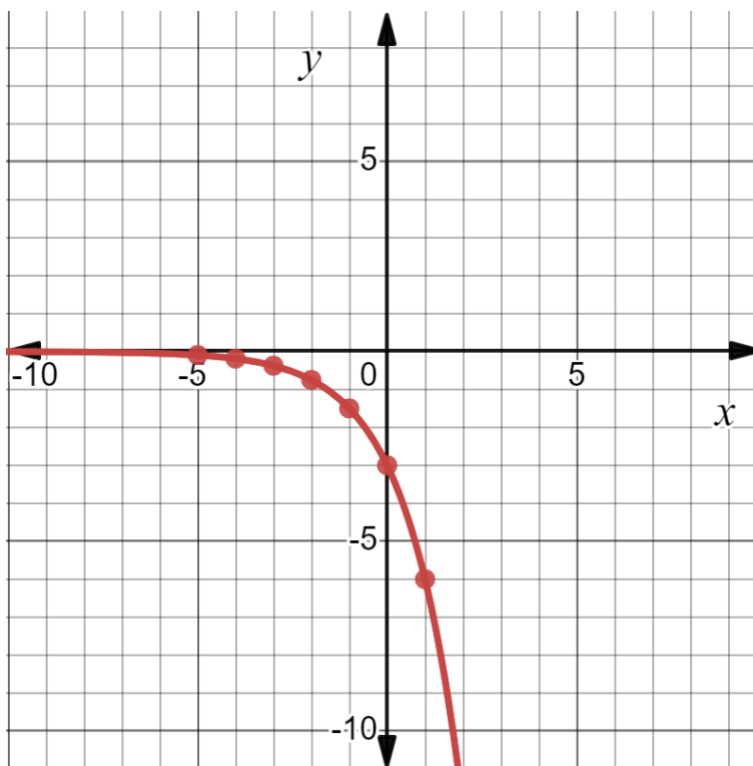
2. For each function, select the box to classify the function as exponential growth or exponential decay.

	$f(x) = -\frac{1}{2}(5)^x$	$g(x) = 4\left(\frac{1}{7}\right)^x$	$f(x) = (2)^x$	$f(x) = (1 + 0.3)^x$
Exponential Decay	☐	☒	☐	☐
Exponential Growth	☒	☐	☒	☒

3. A table of values for an exponential function is shown.

x	-3	-2	-1	0	1	2	3
y	-0.375	-0.75	-1.5	-3	-6	-12	-24

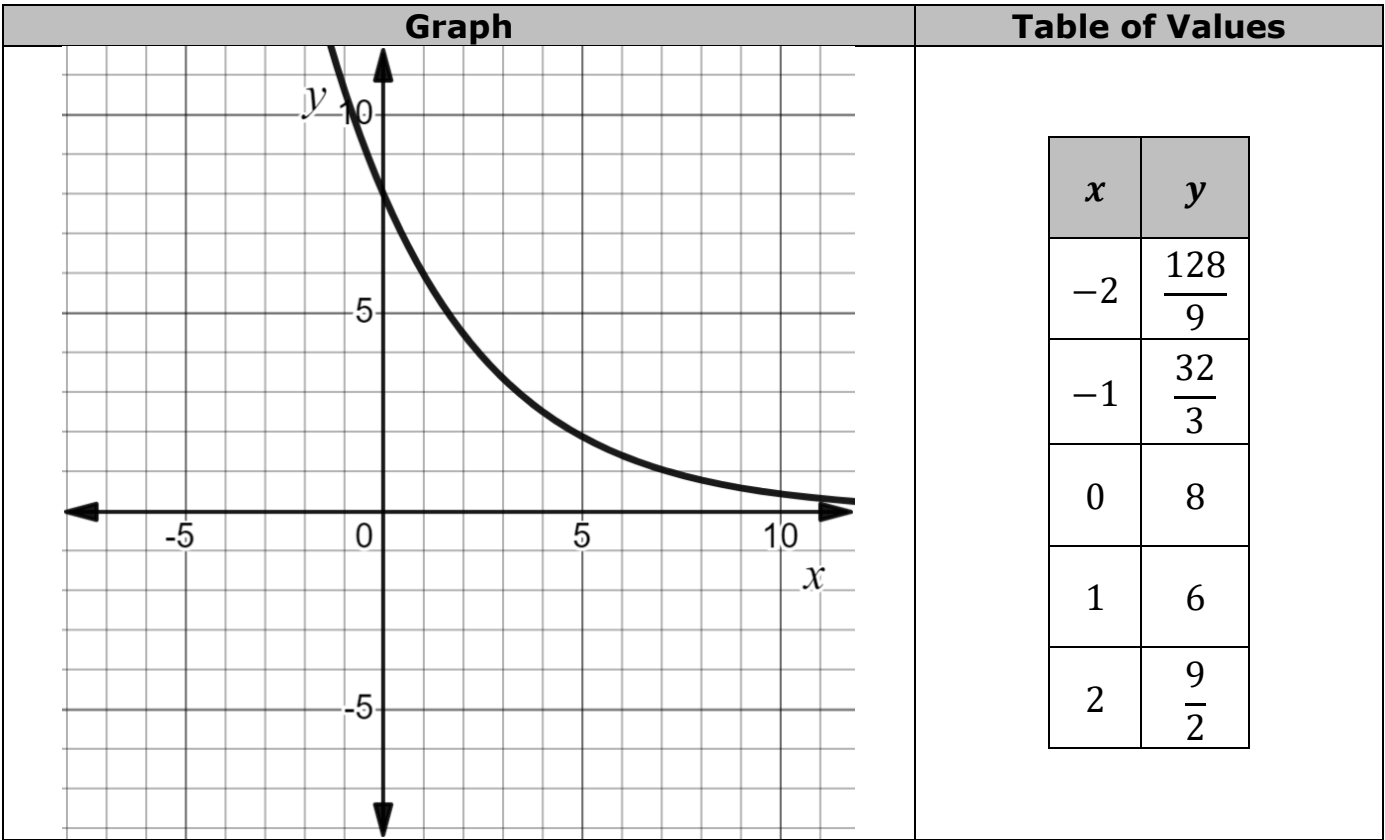
- a. Use the coordinate grid to graph the function.



- b. Write the equation of the exponential function in the form $y = a(1 \pm r)^x$.

Equation	$y = -3(1 + 1)^x$
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4. The graph and table of values for an exponential function are shown.



a. Complete the table.

Key Features			
Domain	$\mathbb{R}$	Range	$y > 0$
y-intercept	(0,8)	Constant Percent Rate of Change	$\frac{3}{4}$
Increasing or Decreasing	decreasing	Positive	$\mathbb{R}$
		Negative	none
Left End Behavior As $x \rightarrow -\infty$,	$y \rightarrow \infty$	Right End Behavior As $x \rightarrow \infty$,	$y \rightarrow 0$

b. Write the equation for the exponential function in the form $y = a(b)^x$.

Equation	$y = 8\left(\frac{3}{4}\right)^x$
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5. The music industry has steadily moved from selling music in a physical format such as records, eight tracks, cassettes, and CDs to selling music in digital formats. In 2001, the music industry sold \$26.5 billion dollars of music in the physical format. Since 2001, the amount of sales of music, in billions of dollars, in the physical format decreased by 10% each year.

Write the function $P(t)$, where P represents the sales, in billions of dollars, of music in the physical format and t represents the number of years since 2001.

Function	$P(t) = 26.5(0.9)^t$
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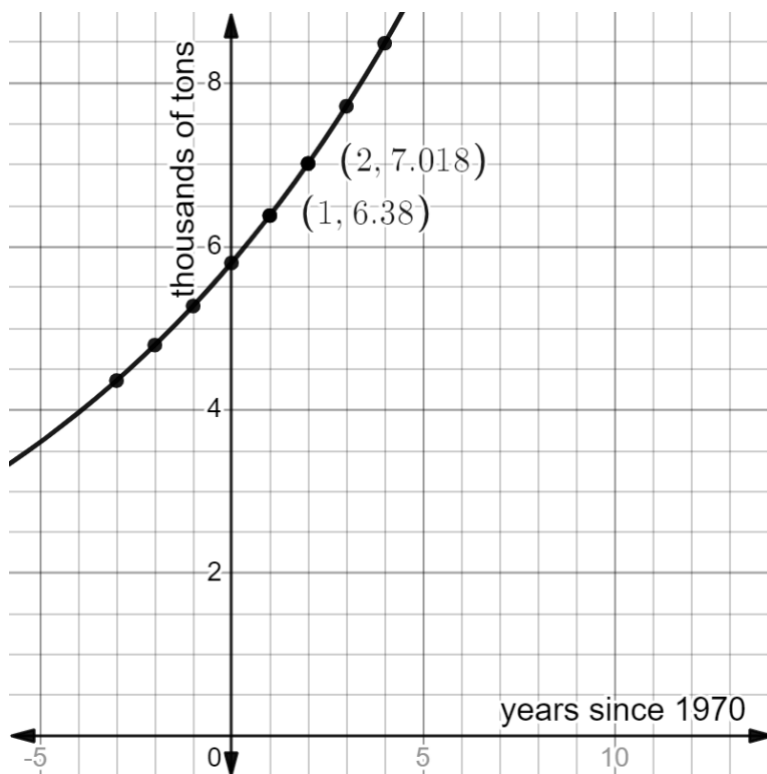
6. As physical formats of music were eliminated, streaming music quickly rose in popularity. Streaming music began earning significant money in 2004 when it earned 0.12 billions of dollars. Money earned from streaming then increased exponentially at a rate of 41.7%.

Write the function $S(t)$, where S represents the amount of money, in billions of dollars, and t represents the number of years since 2004.

Function	$S(t) = 0.12(1.417)^x$
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7. Palm oil production has increased rapidly over the past 50 years. Brazil has become a major producer of palm oil. The graph shows the amount of palm oil that is produced, in thousands of tons, in Brazil. The labeled points on the graph represent the years 1971 and 1972, where x is the number of years since 1970 and y is the amount of palm oil that is produced, in thousands of tons.

Write the equation that models palm oil production, where y represents the amount of palm oil that is produced, in thousands of tons, and x represents the number of years since 1970.



work	
Equation	$y = 5.8(1.1)^x$

8. Lead is toxic to human health and has large impacts on a child's learning and decision-making. Young children who are exposed at a young age can face irreversible effects. Medical professionals have monitored lead concentrations measured in blood samples from children aged one to five years old since 1978. In 1978, the average concentration in children aged one-to-five years old was 29 micrograms per deciliter ($\mu\text{g}/\text{dL}$). Some of the key features of the graph of the function that models the lead concentration in children since 1978 are shown.

Left End Behavior	As $x \rightarrow -\infty, y \rightarrow \infty$	Right End Behavior	As $x \rightarrow \infty, y \rightarrow 0$
Decreasing	$-\infty < x < \infty$	Positive	$-\infty < x < \infty$
Constant Percent Rate of Change	93.5%	Range	$0 < y < \infty$

One point of the function is shown on the coordinate grid. Complete the graph of the function that is described.

